(Please write your Enrollment Number) Enrollment No.

Re-MID-TERM EXAMINATION
(Course Name _B.Tech) (Semester 3rd)
(November, 2023) OFF LINE mode

Subject Code:BMA-205 | **Subject: Thermal Engineering 1** | **Maximum Marks : 30**

Time : 1 ½Hours

Note: Q. 1 is compulsory.

Q1		**(2.5*4)**
	(a)Explain the Kelvin Planck and Clausius statements. How are they violating each other?	
	(b)Explain the PMM1 and PMM2.	
	(c)Define the Zeroth law.	
	(d)Discuss the statement "Entropy is a point function".	
		(5,5)
Q2	**(Attempt any Two Parts) UNIT-1**	
	(a) Define the microscopic and macroscopic point of view with help of examples.	
	(b) Explair the application of first law in nozzle, and turbine.	
	(c)A monoatomic ideal gas (y=1.67) and molecular weight = 40 is compressed adiabatically from 0.1MPa, 300K to 0.2 MPa. The universal gas constant is 0.8314 KJ/KMolK. Calculate the workdone during compression of gas in KJ/Kg.	
		(5,5)
Q3	**(Attempt any Two Parts) UNIT-2**	
	(a) " Entropy is a point function" Justify the statement.	
	(b)Heat flows from a hot reservoir at 900K to another reservoir at 350K. If the entropy change of the overall process is 6.25 KJ/K, Make calculation for the heat flowing out of the high temperature reservoir.	
	(c) Proof the clausius inequality.	